An

Industrial Training Report

on

Enterprise AI Job Market Intelligence (Data Engineering Pipeline)

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in

Department of Computer Science & Engineering



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Session 2026-2027

DECLARATION

I hereby declare that the Industrial Training Report entitled **"Enterprise AI Job Market Intelligence (Data Engineering Pipeline)"** is an authentic record of my own work as part of the requirements of the 45-day Summer Internship Program in Data Engineering with Python, Azure Databricks & Power BI during June 2026. This Industrial Training Report has been submitted as proof of my original work for the fulfillment of the requirement for the award of the degree of B.Tech., Department of Computer Science & Engineering, Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur, under the guidance of Assistant Professor Mr. Shivank Lavania. It has not been submitted anywhere else, under any other program, to the best of my knowledge and belief.

Kartev Vashisth
23ESKCX049

CERTIFICATE



ACKNOWLEDGEMENT

It is my pleasure to express my sincere gratitude to all those who directly or indirectly contributed to the successful completion of my industrial training and this report.

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Finally, I would like to thank my parents and friends for their support and encouragement during the completion of this work.

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ABSTRACT

This report describes the development of an Enterprise AI Job Market Intelligence Platform using Microsoft Azure. The project processes historical and incremental job-market datasets and organizes the data through a Bronze–Silver–Gold Medallion Architecture. Azure Data Lake Storage Gen2 is used to store source CSV files, while Azure Data Factory orchestrates the workflow and can initiate processing through a Storage Event Trigger. Azure Databricks, PySpark, and Delta Lake support data cleaning, validation, deduplication, and incremental upserts. Business-ready Gold tables cover job details, skill demand, company hiring, location hiring, salary summaries, and technology-stack demand. The curated data is loaded into Azure SQL Database using Spark JDBC and visualized through Power BI dashboards. The platform is intended to make job-market information easier to explore by presenting trends in hiring, skills, salaries, work modes, and technology requirements. Any dashboard results should be reported only after validating the actual refreshed data.

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Chapter 1

Introduction

1.1 Background

Online job listings contain useful information about hiring demand, required skills, salary ranges, experience levels, work modes, and locations. When these records are stored in large CSV files, it can be difficult to clean, update, and analyze them consistently. A repeatable data pipeline can convert raw listings into structured information for reporting.

1.2 Problem Statement

Finding useful information from large amounts of job-market data is not easy. It takes time to identify which skills are in demand, which companies are hiring, and what salaries are being offered. Manual processing can also lead to errors, duplicate records, and outdated reports. This project aims to automate the collection, processing, and analysis of job data using Azure and present the results through Power BI dashboards.

1.3 Motivation

The project provides practical experience in cloud data engineering and demonstrates how raw job listings can be transformed into consistent datasets for analysis.

1.4 Objectives

- Build an end-to-end Azure data engineering pipeline for job-market data.
- Ingest historical and incremental CSV files from cloud storage.
- Apply Bronze, Silver, and Gold processing layers using PySpark and Delta Lake.
- Validate records, remove duplicates, and support incremental upserts.

- Load analytical tables into Azure SQL Database.
- Create Power BI dashboards for job trends, skills, salaries, companies, locations, and technology demand.

1.5 Scope of the Topic

The project focuses on batch ingestion, data quality, transformation, analytical aggregation, SQL integration, and dashboard reporting. Predictive models and real-time external job API ingestion are possible future enhancements rather than current features unless separately implemented.

1.6 Organization of the Report

This report introduces the problem and objectives, reviews the technologies, describes the architecture and methodology, discusses implementation and use cases, and concludes with limitations and future scope.

1.7 Technology Overview

The main technologies are Microsoft Azure, Azure Data Lake Storage Gen2, Azure Data Factory, Azure Event Grid Storage Event Trigger, Azure Databricks, Apache Spark, PySpark, Delta Lake, Unity Catalog, Azure SQL Database, JDBC, SQL, and Power BI.

Chapter 2

Literature Review

2.1 Review of Existing Work

Traditional job market analysis workflows generally involve collecting job records, preparing data, applying analytical techniques, and presenting the resulting insights. In a multi-source telecom environment, these activities become more useful when implemented as a repeatable data pipeline rather than as isolated manual operations.

The project adopts a cloud-oriented approach in which raw files are stored centrally, processing is performed using distributed data processing tools, curated data is stored in a relational database, and the results are consumed through a business intelligence dashboard.

2.2 Research Paper Analysis

The supplied project material does not identify specific research papers used as direct inputs to the implementation. Therefore, this report does not attribute particular algorithms or results to external papers. The implementation is based on the technologies and workflow developed during the industrial training.

2.3 Comparative Study (In table format)

2.4 Research Gaps

The current project provides a strong data engineering and analytics foundation, but it does not yet implement machine learning-based churn prediction, real-time streaming, advanced monitoring, or large-scale warehouse analytics. These areas form the basis of the proposed future scope.

Table 2.1: Comparison of manual processing and the implemented pipeline

Aspect	**Manual / Basic Workflow**	**Implemented Workflow**
Data Storage	Individual files	Azure Data Lake Storage Gen2
Processing	Manual / local processing	Databricks with PySpark
Pipeline Control	Manual execution	Azure Data Factory orchestration
Data Layers	No fixed layers	Bronze–Silver–Gold architecture
Data Reliability	File-based management	Delta Lake-based processing
Structured Storage	Local / ad-hoc storage	Azure SQL Database
Visualization	Static analysis	Interactive Power BI dashboard
Triggering	Manual start	Storage Event Trigger

Chapter 3

Methodology

3.1 System Architecture

The platform uses an end-to-end Azure data engineering architecture. Historical and incremental CSV files are stored in Azure Data Lake Storage Gen2. Azure Data Factory orchestrates the workflow, and Azure Databricks executes PySpark transformations across Bronze, Silver, and Gold layers. The Gold tables are loaded into Azure SQL Database and used by Power BI.

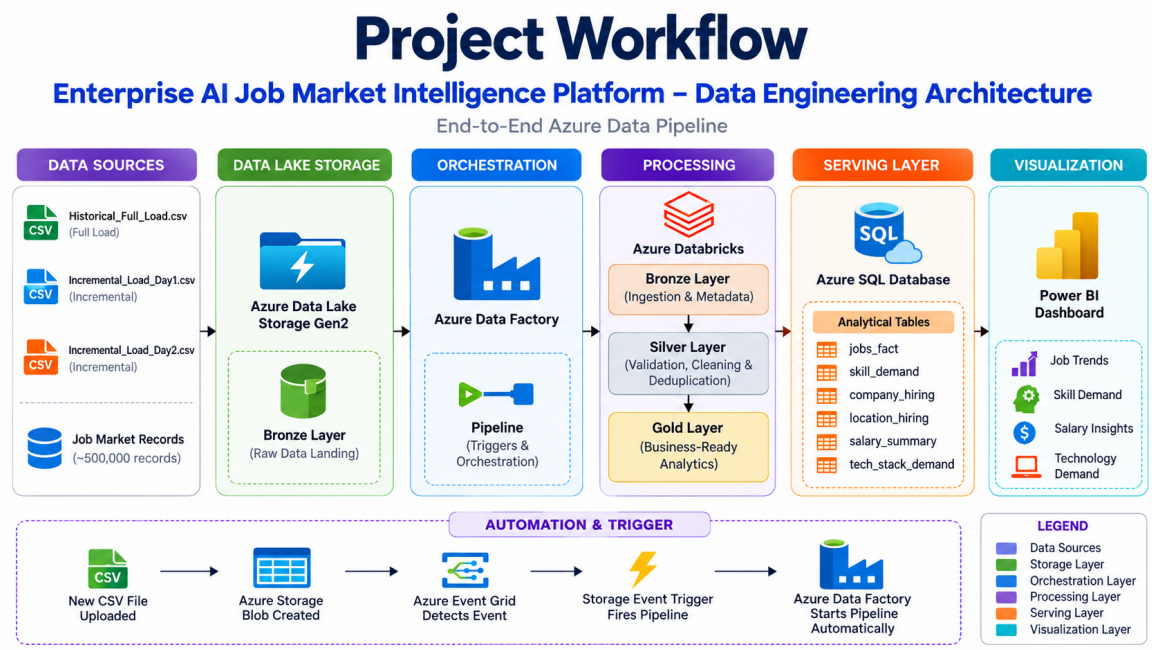


Figure 3.1: Reference architecture for the data engineering platform

3.2 Processing Methodology

1. **Ingestion:** Read historical and incremental job-market CSV files from ADLS Gen2.
2. **Bronze:** Preserve ingested data in Delta format with load metadata, ingestion timestamp,

and source file information.

3. **Silver:** Validate identifiers and important fields, standardize data, remove duplicates, process skill fields, and use Delta Lake MERGE for incremental upserts.

4. **Gold:** Create analytics-ready tables for job facts, skill demand, company hiring, location hiring, salary summaries, and technology demand.

5. **SQL integration:** Write the Gold outputs to Azure SQL Database using Spark JDBC. Skill arrays are converted to strings where required by SQL column types.

6. **Visualization:** Connect Power BI to SQL reporting tables and review the refreshed dashboard.

3.3 Workflow

CSV files → ADLS Gen2 → Event Trigger / ADF → Databricks Bronze–Silver–Gold → Azure SQL → Power BI

3.4 Automation and Refresh

When configured, a Storage Event Trigger can start the Azure Data Factory pipeline after a new file arrives. The SQL refresh procedure should run after Gold transformations and immediately before the SQL loading notebook. Truncate followed by append is not an atomic multi-table refresh; staging tables and transactional publication are recommended for a production-grade implementation.

3.5 Implementation Evidence

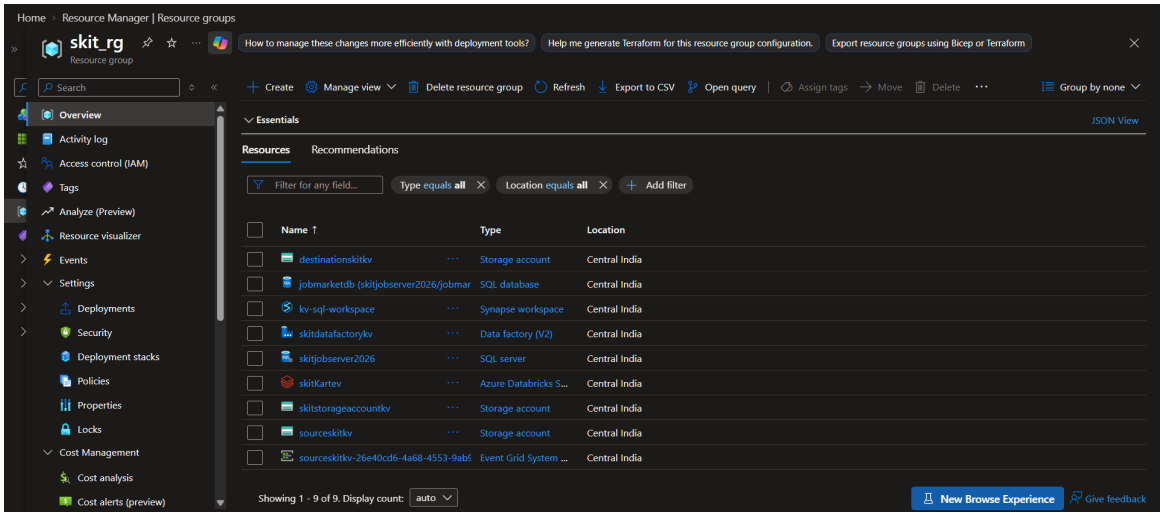


Figure 3.2: Azure resource group used for the project

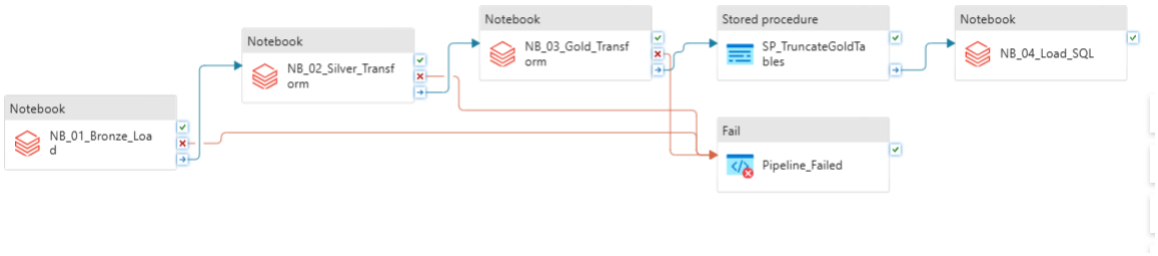


Figure 3.3: ETL pipeline workflow in Azure Data Factory

Chapter 4

Applications and Case Studies

4.1 Applications

The Enterprise AI Job Market Intelligence Platform helps analyze job market data by processing job postings from different datasets through an automated Azure Data Engineering pipeline. The processed data can be used to identify in-demand skills, hiring trends, salary patterns, company hiring activities, and technology demand in the AI industry.

4.2 Industry Use Cases

The project demonstrates several practical Data Engineering use cases:

- Automated ingestion of historical and incremental job market CSV files.
- Distributed data processing using Azure Databricks and PySpark.
- Data cleaning, validation, and deduplication using Bronze–Silver–Gold architecture.
- Incremental data updates using Delta Lake MERGE INTO.
- Storage of analytics-ready data in Azure SQL Database.
- Interactive reporting and visualization using Power BI dashboards.

4.3 Real-World Case Studies

The platform can support recruitment teams, HR analysts, and businesses in understanding changes in the AI job market. For example, the processed data can help identify frequently requested skills, compare salary ranges across experience levels, analyze hiring activity by country, and examine demand for technologies such as Python, SQL, Azure, and Databricks.

The same pipeline architecture can also be adapted to other domains where large datasets need to be ingested, transformed, stored, and analyzed for business intelligence.

4.4 Benefits and Impact

The implemented workflow provides an automated and organized approach to job market data processing. Azure Data Factory and Storage Event Triggers support automated pipeline execution when new files arrive. Azure Databricks, PySpark, and Delta Lake enable data transformation, validation, deduplication, and incremental updates across the Bronze, Silver, and Gold layers.

The Gold tables organize job facts, skill demand, company hiring, location hiring, salary summaries, and technology demand. Azure SQL Database stores the processed analytical data, while Power BI presents the results through interactive dashboards.



```
JobID,JobTitle,CompanyName,CompanyIndustry,CompanySize,CompanyRating,JobDescription,Required
JOBNEW000001,Power BI Developer,Company428 Tech,Retail,51-200,4.6000000000000005,"We are see
JOB472339,Azure Synapse Engineer,Company307 Tech,Manufacturing,10000+,4.5,"We are seeking a
JOBNEW000003,Cloud Data Engineer,Company212 Tech,Energy,501-1000,5.0,"We are seeking a Cloud
JOB147074,Data Integration Engineer,Company453 Tech,Finance,51-200,4.7,"We are seeking a Dat
JOB222973,Data Architect,Company361 Tech,Telecommunications,501-1000,4.6000000000000005,"We
JOB088689,Data Warehouse Engineer,Company316 Tech,Manufacturing,10000+,4.8,"We are seeking a
JOB100476,Business Intelligence Developer,Company180 Tech,Technology,10000+,4.8,"We are seek
JOB219455,AI Engineer,Company348 Tech,Telecommunications,501-1000,5.0,"We are seeking a AI E
JOBNEW000009,Cloud Data Engineer,Company243 Tech,Technology,501-1000,4.9,"We are seeking a C
JOBNEW000010,Data Engineer,Company343 Tech,Manufacturing,201-500,3.3,"We are seeking a Data
JOB481259,Senior Data Analyst,Company312 Tech,Retail,201-500,3.7,"We are seeking a Senior Da
JOB165343,Data Platform Engineer,Company442 Tech,Telecommunications,5001-10000,3.8,"We are s
JOB128896,AI Engineer,Company437 Tech,Manufacturing,1001-5000,5.0,"We are seeking a AI Engin
JOB219296,Analytics Engineer,Company230 Tech,Retail,10000+,3.3,"We are seeking a Analytics E
JOB282086,Data Engineer,Company488 Tech,Technology,201-500,5.0,"We are seeking a Data Engine
JOB013380,Power BI Developer,Company211 Tech,Energy,501-1000,4.6000000000000005,"We are seek
JOB365667,Azure Synapse Engineer,Company200 Tech,Finance,1001-5000,4.8,"We are seeking a Azu
JOB075001,Data Scientist,Company352 Tech,Manufacturing,1001-5000,4.3,"We are seeking a Data
JOB145665,Big Data Engineer,Company378 Tech,Retail,10000+,4.9,"We are seeking a Big Data Eng
JOB282061,Business Intelligence Developer,Company331 Tech,Finance,51-200,3.8,"We are seeking
JOBNEW000021,Data Scientist,Company314 Tech,Telecommunications,201-500,3.5,"We are seeking a
JOB219581,Machine Learning Engineer,Company216 Tech,Manufacturing,51-200,4.2,"We are seeking
JOB417375,AI Engineer,Company429 Tech,Retail,51-200,3.4,"We are seeking a AI Engineer to bui
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Figure 4.1: PySpark transformations for job market data

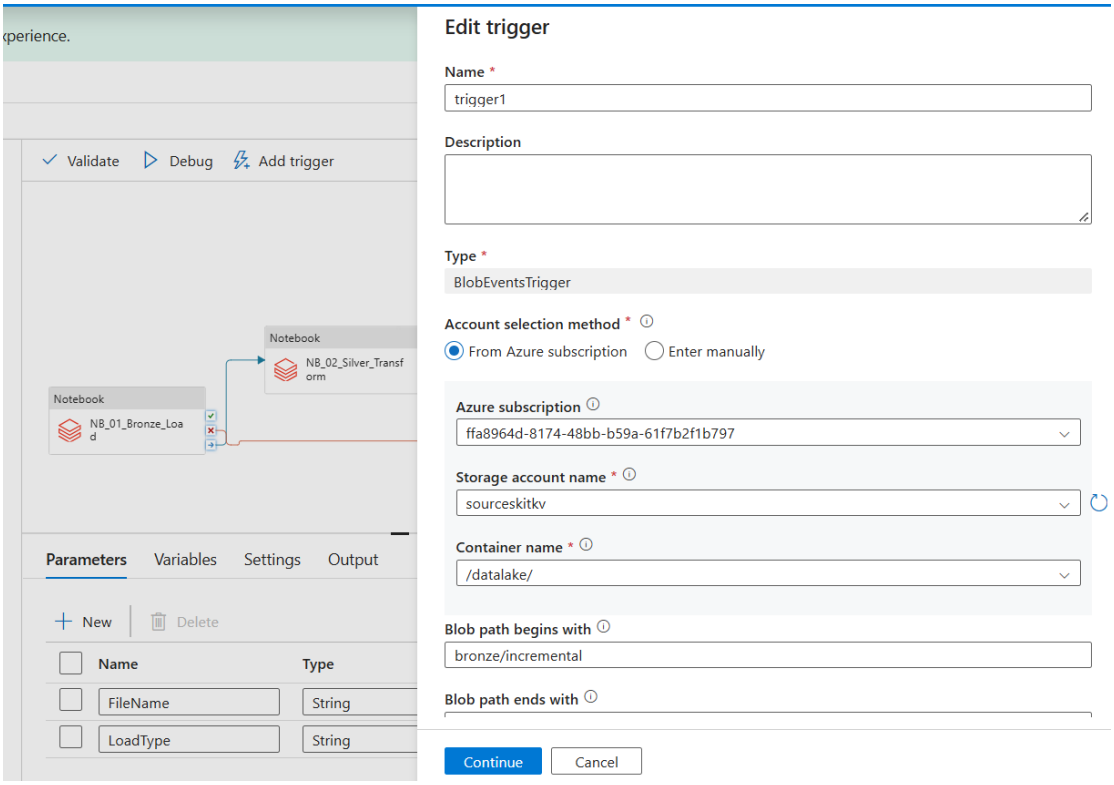


Figure 4.2: Storage Event Trigger configuration for automated data ingestion

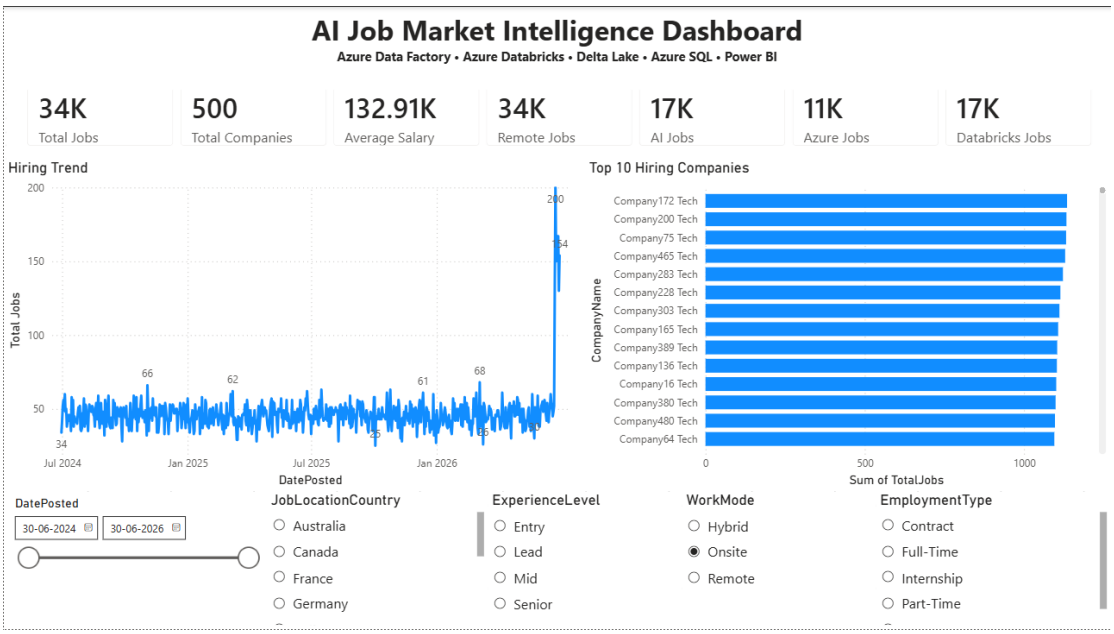


Figure 4.3: Power BI dashboard for AI job market analytics

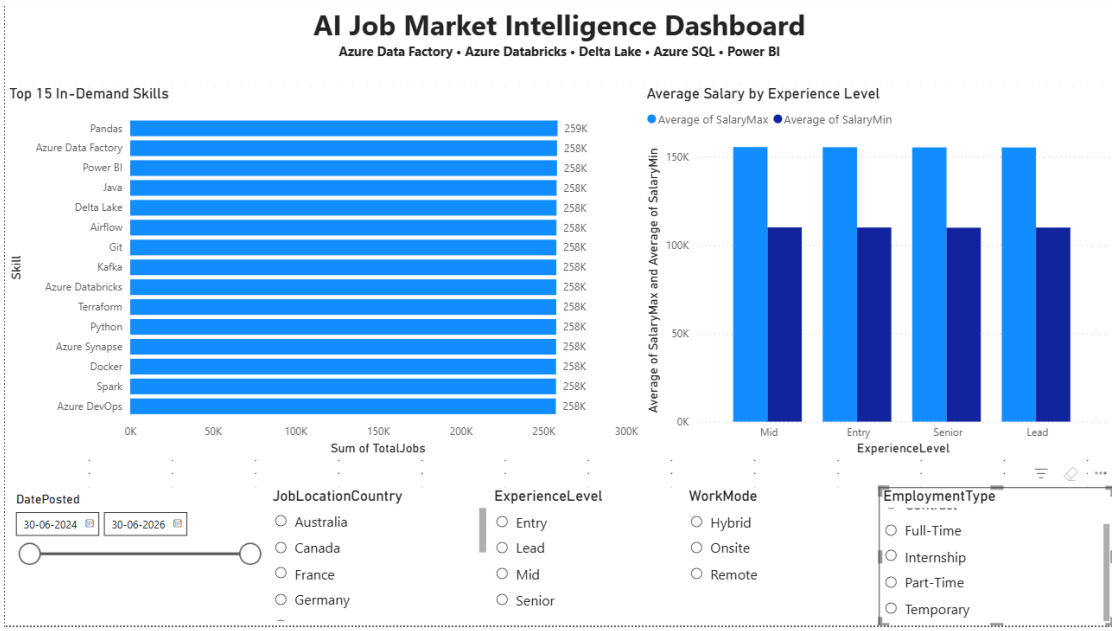


Figure 4.4: Power BI dashboard for AI job market analytics

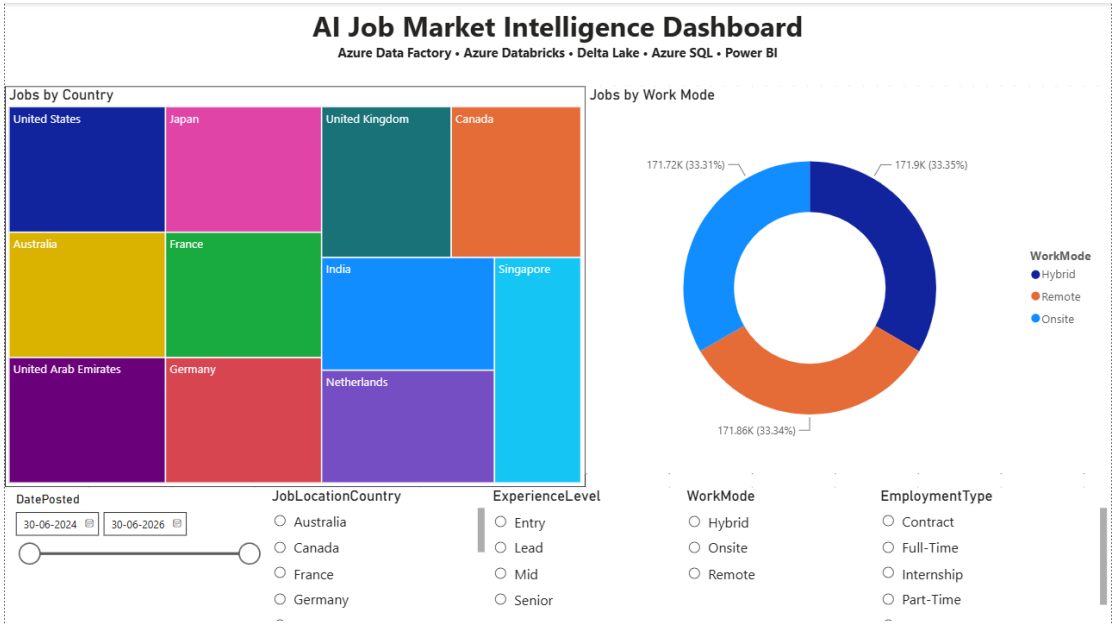
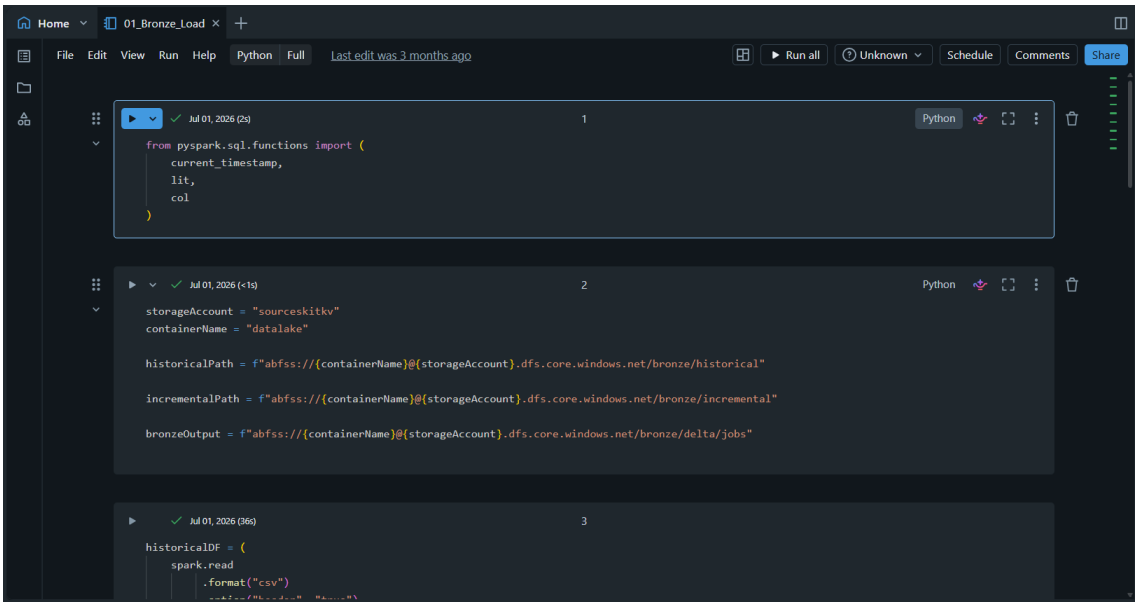


Figure 4.5: Power BI dashboard for AI job market analytics

4.5 Implementation Code Snippets

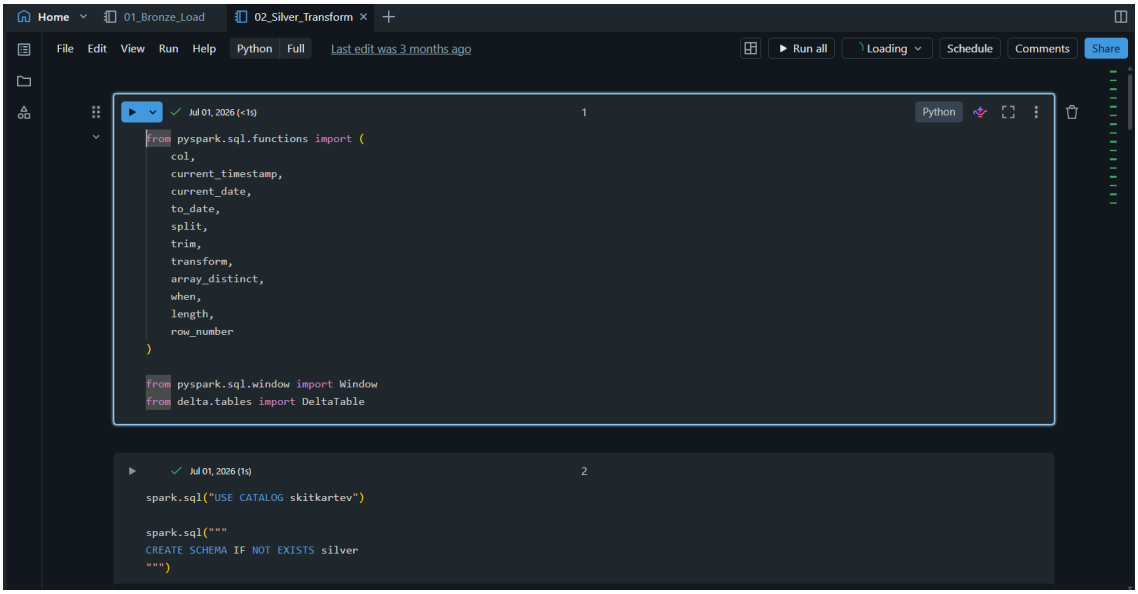
4.5.1 01_Bronze_Load

This notebook ingests historical and incremental job market CSV files and stores the raw data in the Bronze layer with ingestion metadata.



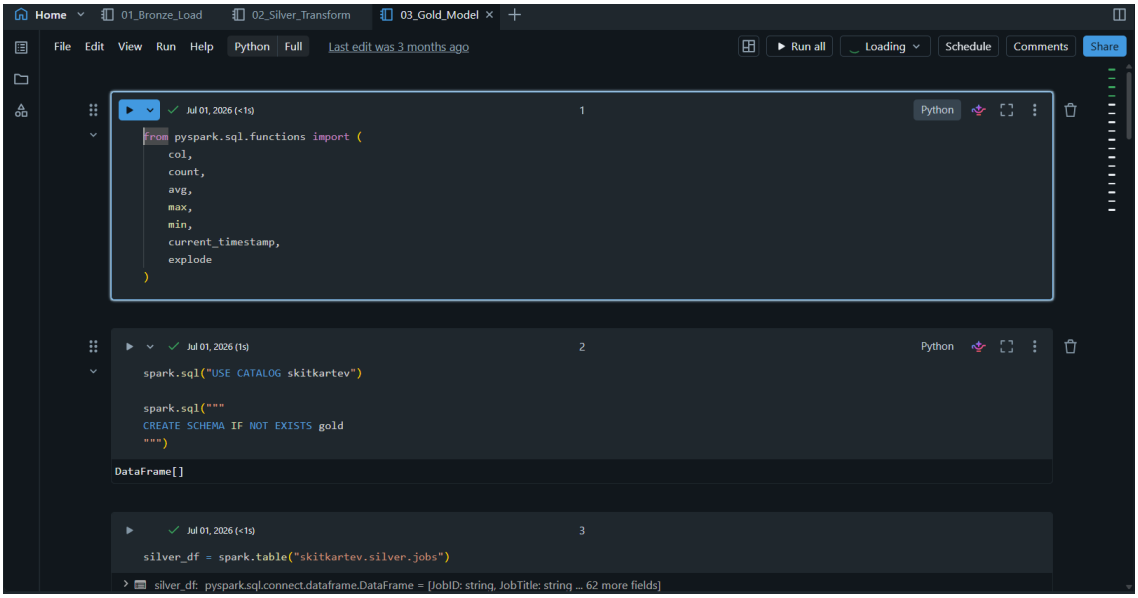
4.5.2 02_Silver_Transform

This notebook validates job records, handles duplicates, transforms skill fields, and applies incremental upserts using Delta Lake.



4.5.3 03_Gold_Transform

This notebook creates analytics-ready Gold tables for job facts, skill demand, company hiring, location hiring, salary summaries, and technology demand.



4.5.4 04_Load_SQL

This notebook loads the six Gold layer tables from Azure Databricks into Azure SQL Database using Spark JDBC. It converts skill array columns into strings for SQL Server compatibility. The loaded data is used in Power BI dashboards to analyze job trends, skill demand, hiring patterns, and salary insights.

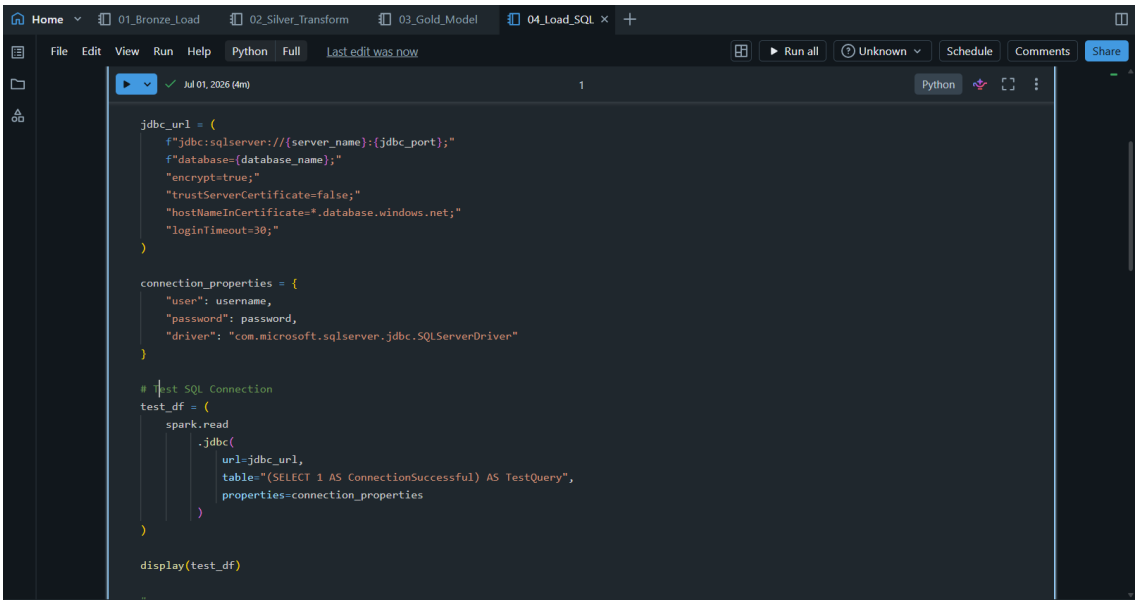


Figure 4.6: Azure Databricks notebook for loading Gold data into Azure SQL Database

Chapter 5

Conclusion

5.1 TAKEAWAYS

The project provided practical experience in developing an end-to-end cloud-based Data Engineering pipeline for AI job market analysis. It integrated data ingestion, transformation, storage, and visualization using Microsoft Azure services.

The main technical takeaways were:

- Understanding cloud data storage using Azure Data Lake Storage Gen2.
- Automating data ingestion and pipeline execution using Azure Data Factory and Storage Event Triggers.
- Processing and transforming job market data using Azure Databricks and PySpark.
- Implementing Bronze–Silver–Gold Medallion Architecture.
- Using Delta Lake for data validation, deduplication, and incremental updates.
- Loading analytical Gold tables into Azure SQL Database using Spark JDBC.
- Developing interactive Power BI dashboards to analyze job trends, skill demand, hiring patterns, and salary insights.

The project also improved practical understanding of Python, SQL, ETL concepts, data quality management, cloud integration, and business intelligence. Overall, it demonstrated how raw job market data can be transformed into useful information for analytical and business decision-making.

5.2 FUTURE SCOPE

The project can be enhanced in the following ways:

1. Integrating external job market APIs to collect updated job postings automatically.
2. Applying Machine Learning models to predict job demand and salary trends.
3. Implementing real-time data processing for faster analysis of newly available job information.
4. Improving data quality monitoring, pipeline logging, error handling, and alerting.
5. Automating Power BI dataset refresh after successful data loading.
6. Enhancing scalability and reliability through optimized data processing and secure cloud infrastructure.

These enhancements can make the platform more scalable and useful for analyzing changes in the AI job market.

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